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作业7.

40.(2). $(\cos\alpha, \cos\beta, \cos\gamma) = (\frac{1}{3}, \frac{2}{3}, \frac{2}{3})$.

$\therefore \frac{\partial u}{\partial x} = \frac{1}{\sqrt{2}}$, $\frac{\partial u}{\partial y} = 0$. $\frac{\partial u}{\partial z} = \frac{1}{\sqrt{2}}$

$\therefore \frac{\partial u}{\partial l} = \frac{\sqrt{2}}{2}$.

41.(2). $\nabla u = 6x\hat{e}_x - 4y\hat{e}_y + 6z\hat{e}_z$.

42.(2). $\nabla u = 2x\hat{e}_x + 2(y-z)\hat{e}_y - 2y\hat{e}_z$.

$\therefore M(-1, 2, 1)$. $\therefore$ 方向为 $\vec{l} = (-2, 2, -4)$

方向导数最小 $\therefore$ 沿方向余弦 $(\frac{1}{\sqrt{6}}, -\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}})$.

$\frac{\partial u}{\partial l} = -2\sqrt{6}$.

43. ∇u 为常矢量 $\Rightarrow \frac{\partial u}{\partial x} = C_1$, $\frac{\partial u}{\partial y} = C_2$, $\frac{\partial u}{\partial z} = C_3$

$\therefore u = C_1 x + C_2 y + C_3 z + C_4$

即: u 为线性函数.

u 为线性函数 $\Rightarrow u = ax + by + cz + d$

$\mathrm{grad}\, u = \frac{\partial u}{\partial x}\hat{e}_x + \frac{\partial u}{\partial y}\hat{e}_y + \frac{\partial u}{\partial z}\hat{e}_z = (a, b, c)$ 为常矢量.

44.(2). $\begin{cases} y\,dy + z\,dz = 0. \\ x\,dx + z\,dz = 0. \end{cases}$ 在 $(1, 1, 3)$ 处.

$dx = dy = -3dz$. $\therefore$ 切向量 $\vec{l} = (3, 3, -1)$

$\therefore$ 切线方程 $\begin{cases} x = 3t + 1 \\ y = 3t + 1 \\ z = -t + 3 \end{cases}$ 法平面方程 $3(x-1) + 3$

46.(2). 法向是 $\hat{n}=(ax_0, by_0, cz_0)$

切平面 $ax_0(x-x_0)+by_0(y-y_0)+cz_0(z-z_0)=0$

法线: $\dfrac{x-x_0}{ax_0}=\dfrac{y-y_0}{by_0}=\dfrac{z-z_0}{cz_0}$.

47. 曲面在 (x_0, y_0, z_0) 处

法向是 $\hat{n}=(6x_0, 2y_0, -2z_0)$

即 $\hat{n}=(3x_0, y_0, -z_0)$.

切平面 $3x_0(x-x_0)+y_0(y-y_0)-z_0(z-z_0)=0$

即: $3x_0x+y_0y-z_0z=3$.

又 ∵ 切平面: $4x+y-z-3+\lambda(x+y-z)=0$.

即 $(4+\lambda)x+(1+\lambda)y-(\lambda+1)z=3$.

$$\therefore \begin{cases} 3x_0=4+\lambda \\ y_0=1+\lambda \\ z_0=1+\lambda \\ 3x_0^2+y_0^2-z_0^2=3 \end{cases} \quad \therefore \begin{cases} x_0=1 \\ y_0=0 \\ z_0=0 \end{cases} \text{ 或 } \begin{cases} x_0=-1 \\ y_0=-6 \\ z_0=-6 \end{cases}$$

即: $x=1$ 或 $x+2y-2z+1=0$.

48. (2) 曲面交点: $ax_0=by_0$

曲面1: 法向是 $\vec{n_1}=(2x_0-a, 2y_0, 2z_0)$.

曲面2: 法向是 $\vec{n_2}=(2x_0, 2y_0-b, 2z_0)$.

$\vec{n_1}\cdot\vec{n_2}=4x_0^2-2ax_0+4y_0^2-2by_0+4z_0^2$.

$=4ax_0-2ax_0-2by_0=0$.

∴ 两曲面相互正交.

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49.(2). 法向量 $\vec{n}=\left(\frac{2}{3}x_0^{-\frac{1}{3}},\ \frac{2}{3}y_0^{-\frac{1}{3}},\ \frac{2}{3}z_0^{-\frac{1}{3}}\right)$

可 $\vec{n}=\left(x_0^{-\frac{1}{3}},\ y_0^{-\frac{1}{3}},\ z_0^{-\frac{1}{3}}\right)$

切平面 $x_0^{-\frac{1}{3}}(x-x_0)+y_0^{-\frac{1}{3}}(y-y_0)+z_0^{-\frac{1}{3}}(z-z_0)=0$

即: $x_0^{-\frac{1}{3}}x+y_0^{-\frac{1}{3}}y+z_0^{-\frac{1}{3}}z=a^{\frac{2}{3}}$.

截距分别为 $a^{\frac{2}{3}}x_0^{\frac{1}{3}},\ a^{\frac{2}{3}}y_0^{\frac{1}{3}},\ a^{\frac{2}{3}}z_0^{\frac{1}{3}}$

$\therefore (a^{\frac{2}{3}}x_0^{\frac{1}{3}})^2+(a^{\frac{2}{3}}y_0^{\frac{1}{3}})^2+(a^{\frac{2}{3}}z_0^{\frac{1}{3}})^2=a^{\frac{4}{3}}\cdot a^{\frac{2}{3}}=a^2$.

3341. $\vec{l}=\left(\frac{1}{2},\ \frac{\sqrt{3}}{2}\right)$.

$\frac{\partial z}{\partial x}=2x,\ \frac{\partial z}{\partial y}=-2y$.

$\therefore \left.\frac{\partial z}{\partial l}\right|_{(1,1)}=1-\sqrt{3}$

3342. $\vec{l}=(\cos\alpha,\ \sin\alpha)$.

$\frac{\partial z}{\partial x}=2x-y=1;\ \frac{\partial z}{\partial y}=2y-x=1$.

$\therefore \frac{\partial z}{\partial l}=\cos\alpha+\sin\alpha=\sqrt{2}\sin\left(\alpha+\frac{\pi}{4}\right)$

(a). $\alpha=\frac{\pi}{4}+2k\pi$ 时 (b). $\alpha=\frac{5}{4}\pi+2k\pi$ 时

(c). $\alpha=-\frac{\pi}{4}+k\pi$ 时 $(k\in Z)$.

3343: $\nabla z=\left(\frac{2x_0}{x_0^2+y_0^2},\ \frac{2y_0}{x_0^2+y_0^2}\right)$

$\therefore \frac{\partial z}{\partial l}=\frac{2\sqrt{x_0^2+y_0^2}}{(x_0^2+y_0^2)}=\frac{2}{\sqrt{x_0^2+y_0^2}}$

3344. $\because \frac{x^2}{a^2}+\frac{y^2}{b^2}=1 \quad \therefore \frac{2x}{a^2}+\frac{2y}{b^2}y'=0 \quad \therefore \frac{dy}{dx}=-\frac{b^2}{a^2}\cdot\frac{x}{y}$

$\left.\frac{dy}{dx}\right|_M=-\frac{b}{a} \quad \therefore \vec{l}=\left(\frac{a}{\sqrt{a^2+b^2}},-\frac{b}{\sqrt{a^2+b^2}}\right)$

$\frac{\partial z}{\partial x}=-\frac{2x}{a^2}=-\frac{\sqrt{2}}{a}, \quad \frac{\partial z}{\partial y}=-\frac{\sqrt{2}}{b}$

$\therefore \frac{\partial z}{\partial l}=\frac{-\sqrt{2}}{\sqrt{a^2+b^2}}+\frac{\sqrt{2}}{\sqrt{a^2+b^2}}=0.$

3345. $\frac{\partial u}{\partial \vec{l}}=\cos\alpha+\cos\beta+\cos\gamma$

$|\nabla u|=\sqrt{3}.$

3346. $u=\frac{1}{\sqrt{x^2+y^2+z^2}}=\frac{1}{r}$

$\nabla u=-\frac{1}{2}r^{-\frac{3}{2}}(2x\hat{e}_x+2y\hat{e}_y+2z\hat{e}_z)$

$=-\frac{1}{(x^2+y^2+z^2)^{\frac{3}{2}}}(x\hat{e}_x+y\hat{e}_y+z\hat{e}_z).$

3347. $\nabla u=2x\hat{e}_x+2y\hat{e}_y-2z\hat{e}_z$

$\nabla u|_A=(2\varepsilon,0,0)$. $\nabla u|_B=(0,2\varepsilon,0)$. 二者之间夹角为90°

3348. ① $|\nabla u|=\sqrt{3}.$

② 令 $r=\sqrt{x^2+y^2+z^2}.$

$\nabla V=(1+1000\pi\frac{x}{r})\hat{e}_x+(1+1000\pi\frac{y}{r})\hat{e}_y+(1+1000\pi\frac{z}{r})\hat{e}_z$

二者的梯度大小至多相差约$10\sqrt{10}\pi$

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3349. $\nabla u = (2ax_0, 2by_0, 2cz_0)$.

$\nabla v = (2ax_0+2m, 2by_0+2n, 2cz_0+2p)$.

$$\cos\theta = \frac{2ax_0(2ax_0+2m)+2by_0(2by_0+2n)+2cz_0(2cz_0+2p)}{\sqrt{(2ax_0)^2+(2by_0)^2+(2cz_0)^2}\cdot\sqrt{(2ax_0+2m)^2+(2by_0+2n)^2+(2cz_0+2p)^2}}$$

$$\lim_{(x_0,y_0,z_0)\to\infty} \cos\theta = \frac{(2ax_0)^2+(2by_0)^2+(2cz_0)^2}{(2ax_0)^2+(2by_0)^2+(2cz_0)^2} = 1.$$

$$\lim_{(x_0,y_0,z_0)\to\infty} \theta = 0.$$

3350. $\frac{\partial u}{\partial l} = f_x\cos\alpha + f_y\cos\beta + f_z\cos\gamma$.

$$\frac{\partial^2 u}{\partial l^2} = (f_{xx}\cos\alpha + f_{yx}\cos\beta + f_{zx}\cos\gamma)\cos\alpha$$
$$+ (f_{xy}\cos\alpha + f_{yy}\cos\beta + f_{zy}\cos\gamma)\cos\beta$$
$$+ (f_{zx}\cos\alpha + f_{zy}\cos\beta + f_{zz}\cos\gamma)\cos\gamma.$$

3351. (a). 由3350

左式 $= \sum(f_x\cos\alpha_i + f_y\cos\beta_i + f_z\cos\gamma_i)^2$

$= \sum(f_x^2\cos^2\alpha_i + f_y^2\cos^2\beta_i + f_z^2\cos^2\gamma_i + 2(f_xf_y\cos\alpha_i\cos\beta_i$
$+ f_yf_z\cos\beta_i\cos\gamma_i + f_zf_x\cos\gamma_i\cos\alpha_i))$

$\because \cos^2\alpha_i + \cos^2\beta_i + \cos^2\gamma_i = 1$

$\cos\alpha_i\cos\alpha_j + \cos\beta_i\cos\beta_j + \cos\gamma_i\cos\gamma_j = 0$,

即: 已证归一基下:

$\sum\cos^2\alpha_i = \sum\cos^2\beta_i = \sum\cos^2\gamma_i = 1$

$\sum\cos\alpha_i\cos\beta_i = \sum\cos\beta_i\cos\gamma_i = \sum\cos\gamma_i\cos\alpha_i = 0$.

$\therefore$ 左 $= f_x^2 + f_y^2 + f_z^2$

即: $\left(\frac{\partial u}{\partial l_1}\right)^2 + \left(\frac{\partial u}{\partial l_2}\right)^2 + \left(\frac{\partial u}{\partial l_3}\right)^2 = \left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 + \left(\frac{\partial u}{\partial z}\right)^2$.

b) $\sum \frac{\partial^2 u}{\partial l_i^2}$

$= f_{xx} \sum \cos^2\alpha_i + f_{yy} \sum \cos^2\beta_i + f_{zz} \sum \cos^2\gamma_i$

$+ 2\left(f_{xy} \sum \cos\alpha_i \cos\beta_i + f_{yz} \sum \cos\beta_i \cos\gamma_i + f_{zx} \sum \cos\gamma_i \cos\alpha_i\right)$

由已交归一基的性质: 右 $= f_{xx} + f_{yy} + f_{zz}$.

$\therefore$ 即: $\frac{\partial^2 u}{\partial l_1^2} + \frac{\partial^2 u}{\partial l_2^2} + \frac{\partial^2 u}{\partial l_3^2} = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}$

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思考题：

1.

1) $\because 0 \le |y| \le \sqrt{x^2+y^2}$

$\therefore 0 \le |y|^3 \le (x^2+y^2)^{\frac{3}{2}}$，记 $r=\sqrt{x^2+y^2}$

$\therefore 0 \le f(x,y) \le r$. $\because \lim\limits_{(x,y)\to(0,0)} r = 0$.

$\therefore \lim\limits_{(x,y)\to(0,0)} f(x,y) = 0. = f(0,0)$

$\therefore f(x,y)$ 在 $(0,0)$ 连续

2) 设任意方向向量 $\vec{l}=(a,b)$，$a^2+b^2=1$

$$\therefore \frac{\partial f}{\partial \vec{l}} = \lim_{t\to 0} \frac{f(0+ta, 0+tb)-f(0,0)}{t}$$

$$\because f(ta,tb) = \frac{(tb)^3}{(ta)^2+(tb)^2} = tb^3 \quad \therefore \lim_{t\to 0} \frac{f(ta,tb)-f(0,0)}{t} = b^3$$

$\therefore \frac{\partial f}{\partial \vec{l}} = b^3$，即任意方向导数均存在且值为 b^3

3) 假如其是可微的，则其全微分形式应为

$df = 0dx + 1dy = dy$. 即 $L(x,y)=y$

若其可微，则须满足

$$\lim_{(x,y)\to(0,0)} \frac{f(x,y)-f(0,0)-L(x,y)}{\sqrt{x^2+y^2}} = 0.$$

此 $= \lim\limits_{(x,y)\to(0,0)} \frac{-x^2y}{(x^2+y^2)^{\frac{3}{2}}}$

设 $y=tx$ $\therefore \frac{x^2y}{(x^2+y^2)^{\frac{3}{2}}} = \frac{t}{(1+t)^{\frac{3}{2}}}$，与 t 有关

$\therefore \lim\limits_{(x,y)\to(0,0)} \dfrac{f(x,y)-f(0,0)-L(x,y)}{\sqrt{x^2+y^2}}$ 不存在.

$\therefore$ 函数不可微，(方向导数 b^3 也不是方向的线性函数.